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Editors
Discrete & Computational Geometry

Dear Editors,

Please consider *Reflexive Descent to Minimal Seeds: Uniform Bounds and Free Sums* as a research article. This is a cover-letter draft for the author's review before an actual submission.

The manuscript studies saturated inclusions of reflexive lattice polytopes. It proves that the reflexive-subpolytope order of a free sum in a direct-sum lattice is the product of the factor orders, giving exact addition of shortest and longest descent lengths and explicit families of increasing depth. The topic concerns lattice polytopes, convex hulls and geometric inclusion orders.

An earlier joint manuscript with Ushashi Bhattacharya is posted at `https://hexagonmath.org/2610.00456v1`. Her contribution is credited in the title footnote and bibliography. The present revision adds the free-sum results, an independently checked planar base, a census-relative interval refinement and a scoped Lean supplement. The exact four-dimensional maximum remains undetermined; the whole manuscript is not formally verified.

The accompanying release contains raw TeX, data, verification scripts and Lean sources with the exact coverage and limitations. It records the assistance used to prepare the proofs and text. Funding and competing interest statements are included in the manuscript.

The author will complete the journal's required declarations in the submission portal after reviewing the final files.

Yours sincerely,

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